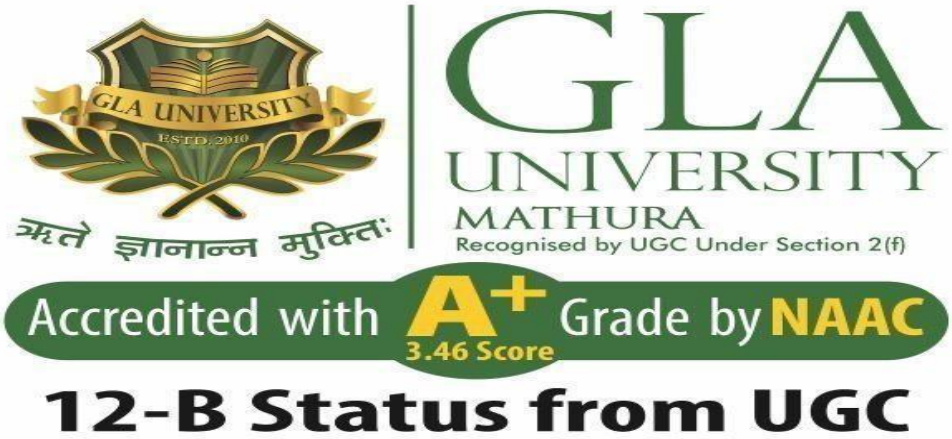
B.Tech CSE (III YEAR – V SEM) (2026-27)

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING & APPLICATIONS



GLA UNIVERSITY

17km Stone, NH-2, Mathura-Delhi Road, P.O. Chaumuhan, Mathura – 281406
(Uttar Pradesh) India

Project Title: CEO-IN-A-BOX: An Autonomous Enterprise Decision-Making Agent

Team Leader: Prakhar Kulshreshtha UR: 2415500343

Team Member 1: Ananya Awasthi UR: 2415500067

Team Member 2: Palak Singh UR: 2415500326

Mentor Name: Govind Gupta

Signature :

Project Synopsis:

CEO-IN-A-BOX: An Autonomous Enterprise Decision-Making Agent

Cover

- **Project title:** CEO-In-A-Box-An Autonomous Enterprise Decision-Making Agent
- **Team name & ID:** (T-140)
- **Institute / Course:** GLA University, Mini-Project
- **Version:** v1.0
- **Date:** 25th Aug 2026

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1. Overview

- **Problem statement:** Modern enterprises generate large volumes of market, financial, and operational data at a pace that exceeds human capacity for timely analysis. Existing decision-support dashboards present data but do not reason over it, and they rarely provide a transparent, explainable rationale for the actions they imply. Decision-makers are thus left to manually synthesize inputs from multiple domains under time pressure, with no mechanism to audit or trace why a particular choice was made
- **Goal:**Build an autonomous multi-agent AI system — AutoCEO — that simulates enterprise decision-making by coordinating five specialized agents. The system analyzes multi-domain signals, produces a final business decision, explains its reasoning at every step.
- **Non-goals:** Real financial transaction execution, connection to live enterprise ERP systems, or institutional-grade quantitative modeling are explicitly out of scope in v1.
- **Value proposition:** A transparent, end-to-end explainable decision pipeline — every agent's reasoning is logged, structured, and displayable — making this demonstrably different from a black-box chatbot and directly addressing the core limitation of existing decision-support tools.

2. Scope and Control

2.1In-scope

- Five-agent orchestration pipeline
  - Structured JSON output validation at each agent boundary
  - Decision logging and explainability audit trail
  - Web-based scenario input dashboard and live decision display
  - Deployment as a live public web app

2.2 Out-of-scope

- Live financial transaction execution
- Real-time ERP integration
- mobile apps,
- advanced portfolio optimization in v1.

2.3 Assumptions

- Users have basic familiarity with business concepts
- Gemini API free tier remains available at current quota levels throughout development
- Synthetic data scenarios are sufficient for demonstrating and evaluating reasoning quality

2.4 Constraints

- All decisions must be explainable
- API usage limits
- Academic project timeline

2.5 Dependencies

- Gemini API (Google AI Studio) — primary LLM provider
- yfinance / Alpha Vantage — optional real market data feeds
- Cloud infrastructure for AI processing

2.6 Acceptance criteria and sign-off

- GIVEN a business scenario CSV WHEN the pipeline runs THEN all five agents produce valid structured JSON and the Executive agent outputs a decision with written rationale within 30 seconds
- GIVEN a crisis event is injected WHEN the pipeline re-runs THEN the Executive decision changes and the change is logged with an explanation
- GIVEN the decision log WHEN the explainability dashboard is opened THEN every agent's input, output, and reasoning trace is visible and queryable

Sign-off table

Stakeholder	Role	Decision area	Signature/Approval	Date
Mr. Govind Gupta	Mentor	Scope, final acceptance	Approved	04/09/2026
Prakhar Kulshreshtha	Product Lead	Release readiness	Approved	

3. Stakeholders and RACI

Activity	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Requirements	Ananya Awasthi	Prakhar Kulshreshtha	Mentor	Team
Design	Palak Singh	Prakhar Kulshreshtha	Mentor	Team
Implementation	Prakhar Kulshreshtha	Prakhar Kulshreshtha	—	Team
Testing	Prakhar Kulshreshtha	Prakhar Kulshreshtha	Mentor	Team
Release	Prakhar Kulshreshtha	Prakhar Kulshreshtha	Mentor	Dept

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4. Team and Roles

Member	Role	Responsibilities	Key skills	Availability	Contact
Prakhar Kulshreshtha	Product Lead	Scope, backlog, reviews	Product, APIs	8 hrs/wk	prakhar.kulshreshtha_cs.aiml24@gla.ac.in
Prakhar Kulshreshtha & Ananya Awasthi	Tech Lead & Backend	Arch, APIs, security	Node, Express, SQL,WebSockets	10 hrs/wk	Ananya.awasthi_cs.aiml24@gla.ac.in
Palak Singh & Ananya Awasthi	Frontend	React UI, state, a11y	React, TS	10 hrs/wk	Palak.singh_cs.aiml24@gla.ac.in

Prakhar Kulshreshtha	QA & Docs	Test plan, E2E, docs	Playwright, writing	8 hrs/wk	Prakhar.kulshreshtha_cs.aiml@gla.ac.in

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5. Week-wise Plan and Assignments

(Example schedule for Sep–Oct 2025; adjust to your calendar.)

Week	Dates	Milestones	Akshya(Lead)	Akshya(Backend)	Manish(Frontend)	Divya(QA/Docs)	Deliverables	Status
1	8-14 Aug	Requirements freeze	Finalize scope	API contracts v0	Wireframes	Test plan v0	Draft SRS	Planned
2	15-24 Aug	Architecture & DB	Arch review	Schema & migrations	UI kit	Test data	ERD, API spec v1	Planned
3	25-31 Aug	Backend scaffolding	Risks, unblock	Auth + Users API	Auth screens	Smoke tests	Auth module	Planned
4	2- 10 Sep	Frontend scaffolding	Sync	Listings API	Routing, forms	Accessibility checks	UI shells	Planned
5	11-17 Sep	Feature set A	Cut scope	Create/Edit listing E2E	Listing pages	CRUD tests	Feature A demo	Planned
6	18-23 Sep	Feature set B	KPIs	Search API	Search UI + Profile	Perf tests	Feature B demo	Planned

7	13-20 Oct	Hardening	Risk burndown	Bug fixes	Bug fixes	Regression suite	Test report	Planned
8	24 Oct-2Nov	Release & deck	Final sign-off	Release notes	Polish UI	User manual	v1.0, slides	Planned

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6. Users and UX

6.1 Personas

- **Business Student / Evaluator:** Needs to see a clear, readable decision with an explanation they can follow and question during a viva
- **System Tester:** Wants to inject different scenario data and verify that agent outputs and the final decision change appropriately

6.2 Top User Journeys

User opens dashboard → selects or uploads a business scenario (market/finance/HR data) → clicks 'Run' → watches five agents run sequentially → reads the Executive Agent's final decision with rationale → views the full decision log for explainability → optionally injects a crisis and re-runs to see adaptive re-planning

6.3 User Stories

As an evaluator, I want to see each agent's reasoning step so that I can verify the system is not a black box.

6.4 Accessibility & Localization

- Simple language explanations
- Mobile-friendly interface

- English language support

7. Market and Competitors

7.1 Competitor table

Competitor	Product	Target users	Key features	Pricing	Strengths	Weaknesses	Our differentiator
AutoGen (Microsoft)		Researchers, developers	Multi-agent conversations, debate-style reasoning	Freemium (Pro: $14.95/mo)	Professional-grade tools, 50M+ users, real-time data, strong community	No explicit decision hierarchy; agents debate freely, hard to control output structur	Fixed hierarchy: 4 specialists report to 1 executive; structured JSON at every step
CrewAI		developers	Role-based agents, quick to prototype	Freemium (Premium: $239/yr)	Expert opinions, comprehensive data, dividend focus	Framework controls flow, hard to explain or audit exact routing logic in a viva	LangGraph: explicit graph = every routing decision is visible and explainable
ChatGPT / Gemini (vanilla)		General Users	Conversational AI, broad knowledge	Free (Gold: $5/mo)	Easy execution, gamified UX, popular with young users	No persistent state, no structured multi-agent pipeline, no decision logging	Persistent decision log, Pydantic-validated outputs, full audit trail

7.2 Positioning

AutoCEO's differentiator is the combination of structured multi-agent hierarchy + enforced schema validation + complete explainability logging. Each of these exists in isolation in other tools, but no free-tier student-accessible system combines all three in a single end-to-end pipeline.

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8. Objectives and Success Metrics

- O1: Pipeline completeness — all 5 agents run end-to-end on a scenario and produce valid JSON — by Week 6
- O2: Explainability — 100% of decisions have a logged, queryable reasoning trace — by Week 7
- O3 Adaptability — system produces a measurably different decision when crisis data is injected — by Week 8
- O4: Deployment — live public URL accessible without local setup

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9. Key Features

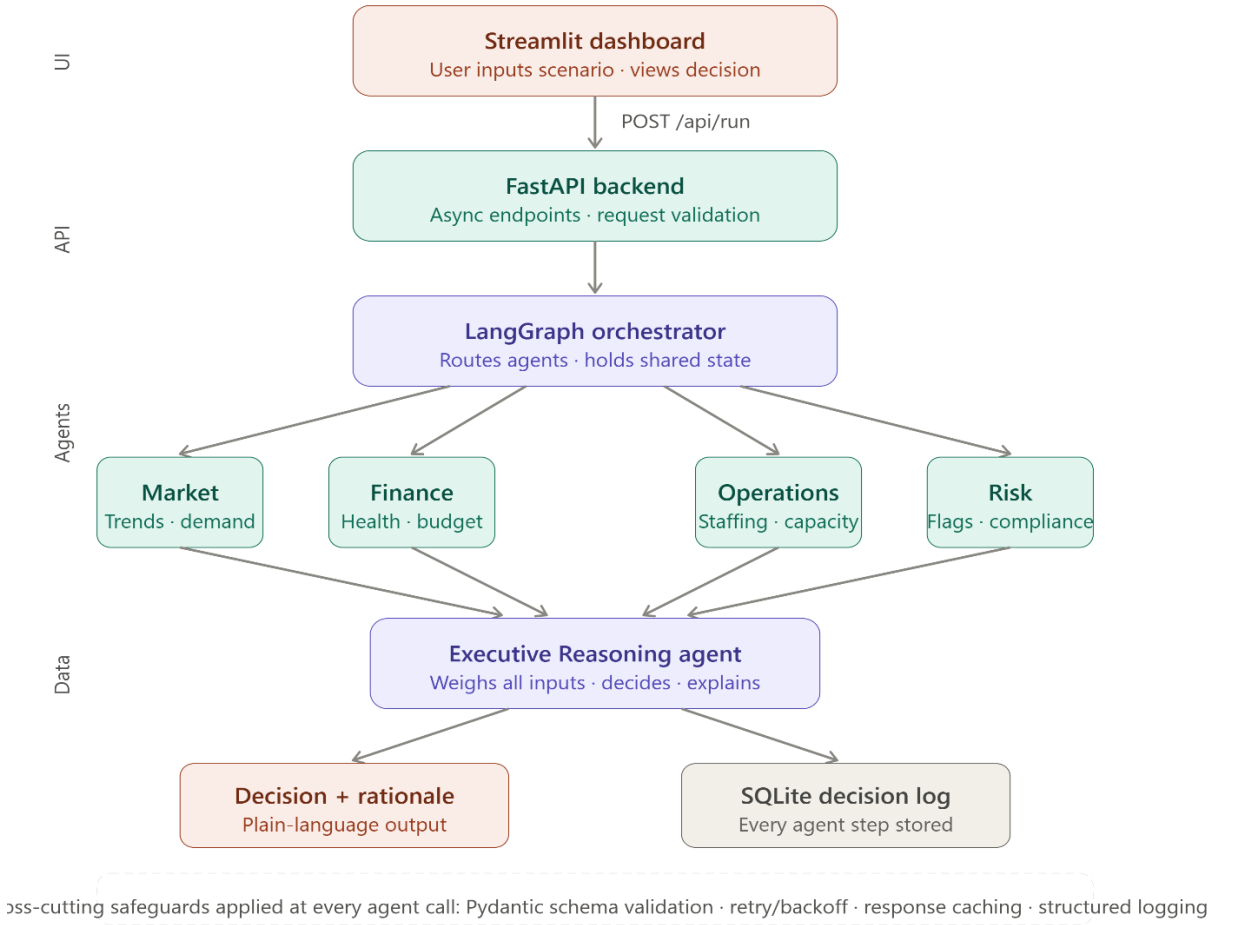
Feature	Description	Priority	Dependencies	Acceptance criteria

Market Intelligence Agent	Reads market/sector data, identifies top trend, outputs demand direction and confidence score as JSON	Must	SMTP	GIVEN market CSV WHEN agent runs THEN returns valid JSON with trend_summary, demand_direction, confidence, recommendation within 10s
Finance Agent	Assesses financial health from budget/revenue data, recommends safe spend range	Must	Auth, Storage	GIVEN finance CSV + market summary WHEN agent runs THEN returns financial_health, safe_budget_range, and budget_flags
Operations Agen	Evaluates team capacity and staffing from HR roster, recommends hire/fire/reassig	Must	Listings	GIVEN HR CSV WHEN agent runs THEN returns capacity_status, staffing_recommendation, operational_risk
Risk & Compliance Agent	Flags any recommendation that violates predefined compliance rules or introduces unacceptable risk	Must	Auth	GIVEN all prior outputs WHEN compliance rules are applied THEN returns risk_level and any violations

Executive Reasoning Agent	Synthesizes all four agent outputs into a single decision with written rationale and re-evaluation trigge	Must	Auth	GIVEN all 4 agent outputs WHEN executive runs THEN returns decision, rationale, key_influences, risk_accepted, reevaluation_trigger

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10. Architecture



10.1 High-Level Architecture

The system follows a modular and scalable architecture designed to support realtime data processing and personalized AI-driven advisory.

- **Clients:**

Web-based Single Page Application (SPA) developed using React. The client provides dashboards, charts, and a conversational interface for user interaction.

- **Backend Services:**
  - **Authentication Service:** Handles user registration, login, and secure access.
  - **Profile Service:** Manages user risk profiles, investment goals, and preferences.
  - **Market Analysis Service:** Fetches and processes real-time and historical market data.
  - **Recommendation Engine:** Generates personalized buy/hold/sell suggestions using AI models.
- **Data Stores:**
  - **MySQL Database:** Stores user data, investment profiles, portfolios, and logs.
  - **Object Storage:** Stores charts, reports, and backtesting results if required.
- **Integrations:**
  - Market Data APIs (Yahoo Finance / Alpha Vantage)
  - LLM Provider (OpenAI / Groq)
  - SMTP service for alerts and notifications

10.2 API spec snapshot

Endpoint	Method	Auth	Purpose	Request schema	Response schema	Codes

/api/auth/register	POST	—	Create account	Run full 5-agent pipeline on scenario data	201 {id}	201, 400
/api/profile/create	POST	JWT	Create investment profile	riskLevel, goals	201 {listingId}	201, 400, 401
/api/market/analyze	GET	-	Analyze stock	Fetch full decision log for a past run	200 {items[], total}	200

10.3 Configuration and Secrets

- All API keys stored in .env file — never hardcoded in source
- .env added to .gitignore before first commit — verified before pushing to GitHub
- Gemini API key rotated if accidentally exposed

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11. Data Design

11.1 Core Entities

- Scenario
- AgentCall
- DecisionRun
- ComplianceRule

User data is securely stored with encryption and strict access control.

User data is securely stored with encryption and access control.

11.1 Data dictionary

Entity	Field	Type	Null?	Allowed values	Source	Notes
Decision Run	run_id	UUID	No	—	System	PK
Decision Run	scenario_id	String	No		User	Unique
Decision Run	final_decision	Enum	No	Low/Medium/High	User	PK
AgentCall	call_id	UUID	No		Market API	Indexed
AgentCall	Run_id	UUID	No	≥ 0	System	—

11.2 Schemas and Migrations

- ER Diagram included in appendix
- Database migrations version-controlled
- Rollback tested on staging environment

11.3 Privacy, Retention, Backup, and DR

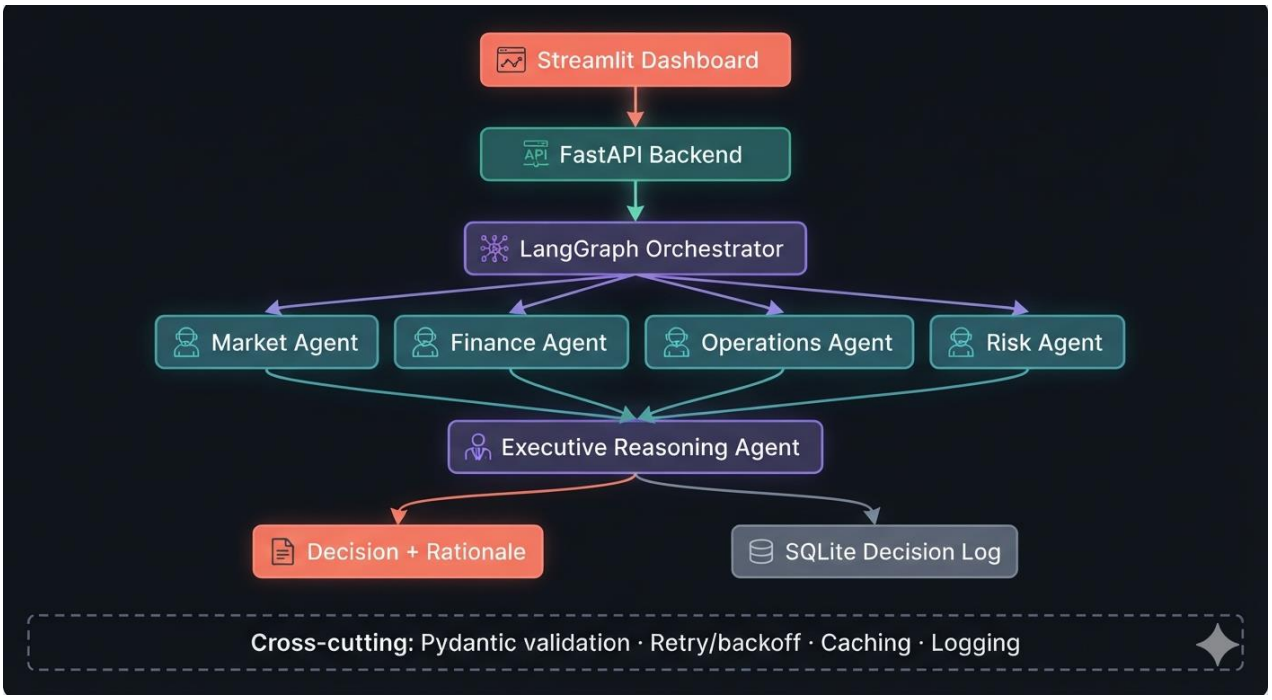
- Personally Identifiable Information (PII): name, email

- Retention policy: delete inactive accounts after 12 months
- Nightly automated backups
- **RTO:** 4 hours
- **RPO:** 24 hours

12. Technical Workflow Diagrams

The following diagrams are included to explain system behavior:

- State Transition Diagram
- Sequence Diagram
- Use Case Diagram
- Data Flow Diagram (DFD)
- Entity Relationship Diagram (ERD)
- System Architecture Diagram



13. Quality: NFRs and Testing

13.1 Non-functional requirements

Metric	SLI	Target (SLO)	Measurement
Availability	Uptime %	≥ 99.0%	Uptime monitor
Latency	p95	≤ 1000 ms	APM
Error rate	5xx %	≤ 1%	Logs
Security	Critical CVEs	0	Vulnerability scanner

13.2 Test plan

Area	Type	Tools	Owner	Coverage target	Exit criteria
Backend	Unit	Jest	Prakhar	70%	No P1/P2 defects
UI	E2E	Playwright	Ananya	60%	Pass rate ≥ 95%
API	Integration	Postmann	Palak	All Scenarios	All critical pass

13.3 Environments

- Development → Staging → Production
- Feature flags used for experimental features

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14. Security and Compliance

14.1 Threat model (STRIDE)

Asset	Threat	STRIDE	Impact	Likelihood	Mitigation	Owner
Auth tokens	Theft	Spoofing	High	Medium	HTTPS, short TTL, rotation	Rohit
User data	SQLi	Tampering	High	Low	Parameterized queries, WAF	Rohit

14.2 AuthN/AuthZ

- Email + password authentication

- JWT-based authorization
- Role-based access control

14.3 Audit and logging

- Log signups, logins, listing CRUD, reports; 90-day retention.

14.4 Compliance

- Academic project; follow institute policy; no third-party data sharing.

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15. Delivery and Operations

15.1 Release plan

- Version v1.0 demo at project submission
- Incremental feature rollout

15.2 CI/CD and rollback

- CI pipeline: lint → test → build → deploy
- Rollback using previous container image

15.3 Monitoring and alerting

Metric	Threshold	Alert to	Runbook
p95 latency	> 1200 ms	Tech Lead	“API Latency” runbook
Error rate	> 2%	Tech Lead	“Error Spike” runbook

15.4 Runbooks

- **API Latency:** check DB indexes → scale pods → revert change.

- **Error Spike:** inspect logs → roll back → create incident note.

15.5 Communication plan

- Standups Mon/Wed/Fri. Weekly status to mentor each Friday. Bi-weekly demo.

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16. Risks and Mitigations

16.1 Risk heatmap

Risk	Probability	Impact	Score	Mitigation	Owner	Status
Gemini API free-tier quota exhausted during crunch	Medium	High	12	Implement response caching; use Groq as backup LLM	Prakhar Kulshreshtha	Open
Schedule slip due to LangGraph learning curve	Medium	Medium	9	Build plain-Python pipeline first (Week 1–2), migrate to LangGraph	Prakhar Kulshreshtha	Open
Prompt quality insufficient for coherent decisions	Low	Medium	6	Hand-test all prompts in AI Studio before coding; iterate until JSON is consistent	Prakhar Kulshreshtha	Open

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17. Research and Evaluation

- **17.1 Study of Existing Platforms**
  Existing multi-agent frameworks (AutoGen, CrewAI, vanilla LangChain agents) were evaluated against three criteria: explainability of routing, suitability for a fixed decision hierarchy, and

free-tier accessibility. LangGraph was selected because it is the only framework that models the pipeline as an explicit, inspectable graph rather than hiding routing behind framework abstractions — a critical requirement for a project where viva evaluators will ask exactly how data flows between agents..

- **17.2 Evaluation Using Historical Data**

  AutoCEO will be evaluated against 5–10 hand-crafted synthetic business scenarios covering: normal operations (baseline), demand contraction (market shock), budget overrun (finance shock), team under-capacity (operational shock), and compliance violation (risk shock). For each scenario, the Executive agent's decision is manually reviewed for coherence, risk balance, and alignment with the injected conditions.

- **17.3 User Feedback**

  Peer review from classmates and faculty mentor will be used to assess clarity of the decision explanations and usability of the dashboard

- **17.4 KPI Tracking**

  Key metrics include response time (p95), recommendation accuracy, user engagement, and system uptime. Regular monitoring ensures reliability and continuous improvement.

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18. Appendices

18.1 Glossary

- **AI:** Systems simulating human intelligence.
- **FastAPI:** A modern Python web framework for building
- **Pydantic:** A Python library for data validation

18.2 References

- **LangGraph documentation**: https://langchain-ai.github.io/langgraph/
- **Google AI Studio (Gemini API):** https://aistudio.google.com
- **FastAPI documentation:** https://fastapi.tiangolo.com
- **Streamlit documentation:** https://docs.streamlit.io
- **Pydantic documentation:** https://docs.pydantic.dev
- **yfinance library:** https://github.com/ranaroussi/yfinance
- **IEEE:** "Multi-Agent Systems for Decision Support" – https://ieeexplore.ieee.org